4.2 Adding, Subtracting, and Multiplying Polynomials

Name _____ Period ____ Date _____

Algebra II 4.2 · Student sheet

I can…

- add and subtract polynomials.
- multiply polynomials, including three binomials.
- use special product patterns and Pascal's Triangle.

Vocabulary · fill in the missing word

Like terms — terms with the same _____ raised to the same powers

Square of a binomial — $(a \pm b)^2 =$ _____

Sum and difference — $(a + b)(a - b) =$ _____

Cube of a binomial — $(a \pm b)^3 =$ _____

Pascal's Triangle — each number is the sum of the two above it; row n gives the _____ of $(a + b)^n$

Part 1 · Pascal's Triangle

Build Pascal's Triangle through row 6 (each number is the sum of the two above it).

Two patterns you notice: _____

$(x + 1)^2 =$ _____ $(x + 1)^3 =$ _____

Part 2 · Notes and examples (fill in each note, then work the example)

Subtracting: To subtract, distribute the minus sign to _____ term.

Example 1

Find the difference: $(3x^3 - 2x^2 + 5) - (x^3 + 4x^2 - x + 2)$

Example 2

Find the product: $(x - 2)(3x^2 + x - 4)$

Example 3

Find the product: $(x + 1)(x - 3)(x + 2)$

Cube of a binomial: $(a + b)^3 =$ _____

Example 4

Find the product: $(2x - 3)^3$

Pascal's Triangle: Row n of Pascal's Triangle gives the coefficients of _____.

Example 5

Use Pascal's Triangle to expand $(x + 2)^4$.

Part 3 · Partner practice

Partner 1

Find the sum: $(x^2 + 3x - 1) + (4x^2 - 3x + 6)$

Partner 2

Find the product: $(x + 3)(x - 1)(x - 2)$